MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS) REGULAR EXAMINATION NOVEMBER 2025
FIRST SEMESTER
Multi-Disciplinary Courses (MDC) - MG1MDCAQC100 - GENERAL PERSPECTIVES IN AQUACULTURE
(2024 ADMISSION ONWARDS)
Q paper Code : 25900730
Maximum Marks : 35
Part A
Very Short Answer Questions
Answer any 10 out of 12 Questions
Each Question carries 2 marks

1. Explain the term 'stenohaline' and give an example of an organism that falls under this category.

Stenohaline organisms are those that can tolerate only a narrow range of salinity. They cannot survive large fluctuations in salt concentration and are therefore restricted to environments where salinity remains relatively stable.

Example:Goldfish, Corals or any example for stenohaline species

2. Define fish marketing
Fish marketing refers to the process of moving fish and fishery products from producers to consumers through a series of activities such as handling, processing, packaging, transportation, storage, pricing, and distribution. It ensures that fish reaches the right consumers in the right form, at the right place, time, and price, while maintaining quality and safety.
3. What is ambient temperature?
Ambient temperature is the temperature of the surrounding environment or the air in the immediate area where an object or organism is located. It represents the natural, existing temperature without any artificial heating or cooling.
4. Name 2 egg laying ornamenral fish.
Gold fish, Angel fish, Gourami, Fighter fish or any other egg laying ornamental fish
5. What are the main objectives of monosex aquaculture?

To achieve uniform and faster growth, To prevent unwanted breeding, To ensure better feed conversion and size uniformity, To improve overall yield and profitability

6. Explain why the common carp lost its cultural prominence in China during the 6th century AD, and identify the alternative species that gained popularity

The Chinese emperor at that time belonged to the Li family.

The Chinese word for carp, “Li”, has the same pronunciation as the emperor’s surname.

Out of respect and to avoid the symbolic act of “killing the emperor,” people were discouraged or even forbidden from harvesting or eating common carp.

As a result, the fish lost its dominant position in culture and aquaculture.

Species that gained popularity : Black carp, Grass carp, Silver carp Bighead carp

7. Define pen culture and its importance.

Pen culture is an aquaculture system in which fish or other aquatic organisms are reared in enclosures constructed with fixed nets or screens placed in natural water bodies such as lakes, reservoirs, or rivers. The bottom is natural, while the sides are fenced to confine the stock.

Importance

Low-cost and easy to install compared to cages or ponds.

Utilizes natural water bodies efficiently without major environmental alteration.

Provides better growth since organisms benefit from natural food available in the ecosystem.

Improves fish production and rural livelihoods by increasing yield from open-water fisheries.

8. Which aquaculture systems produce the highest and lowest yield?

Recirculating Aquaculture Systems (RAS) and intensive pond/cage culture produce the highest yields

Lowest yield: Extensive aquaculture systems

9. What are the ideal levels of dissolved oxygen in water?

Ideal / Optimal: 5–8 mg/L, Most cultured fish grow best when DO is kept above 5 mg/L.

10. What is habitat conservation?

Habitat conservation is the protection, management, and restoration of natural environments to ensure that plant and animal species can survive, reproduce, and maintain healthy populations. It focuses on preserving the ecological conditions and resources, necessary for the long-term sustainability of biodiversity.

11. Describe the ideal temperature range for a tropical aquarium. How to maintain it?

The ideal temperature range for a tropical aquarium is typically 24–28°C

To Maintain This Temperature

Use an Aquarium Heater, Monitor with a Thermometer, Avoid Direct Sunlight, Use a Tank Cover or Lid:

For Cooling : Use an aquarium fan or chiller, Increase aeration

12. Scientific name of Miss Kerala and Malabar puffer fish is

Miss Kerala : *Sahyadria denisonii (Puntius denisonii)*

Malabar puffer fish : *Carinotetraodon travancoricus*

Part B
Short Answer Questions
Answer any 3 Questions
Each Question carries 5 marks

13. Discuss the key revenue sources in aquaculture and how they affect profitability

1. Sale of Market-size Fish or Shellfish

This is the primary revenue source.

Higher survival, faster growth, and better feed conversion increase harvest volume.

Good size and quality fetch premium prices, boosting profit.

2. Seed/Fry/Fingerling Production

Demand for quality seed is high, giving strong margins.

Reliable seed production reduces production risk and stabilises income.

3. Value-added Products

Examples: fillets, smoked fish, pickles, dry fish, ready-to-cook items.

Adds 10–50% more value compared to raw fish.

Reduces post-harvest loss and widens the market.

4. By-products

Includes fish waste, scales, skin, viscera used for Fish meal, Fish oil, Fertilizers, Collagen/gelatin
These generate additional income with minimal extra cost.

5. Integrated Aquaculture Systems

Revenue can also come from: Rice–fish farming, Fish–poultry or fish–livestock systems, Aquaponics. These increase overall farm productivity and reduce input costs.

6. Ornamental Fish and Aquarium Trade

High-value species, broodstock sales, live plants, and aquarium setups offer:

High profit per unit volume

Strong demand in domestic & export markets

7. Licensing, Tourism, and Recreational Fishing

Some farms earn from, Farm visits, Educational tours, Sport/recreational fishing

14. Describe the setup of a longline culture system.

A longline culture system is a type of open-water mariculture setup used mainly for mussels, oysters and seaweeds. It consists of a horizontal main line floating on the water surface and anchored securely at both ends.

Setup of a Longline Culture System : Main Rope/Longline, Floats/Buoys, Anchors and Mooring System, Culture Units

Site Requirements:Clean, well-flushed water with moderate currents.

Adequate depth to keep droppers from touching the bottom.

Function: The longline floats on the surface while the culture organisms grow on vertical ropes below, benefiting from natural food and good water circulation.

15.What innovative material would you use for aquarium decor to enhance both aesthetics and fish comfort?

An excellent innovative material for aquarium décor that enhances both aesthetics and fish comfort is:

3D-printed, aquarium-safe bioplastic structures

Customizable designs, Fish comfort, Safe and inert, Lightweight yet durable, Eco-friendly

Alternative innovative materials

Silicone-based soft structures, Coconut fiber bio-decor, Recycled glass Terracotta ceramics for breeding shelters

16.How does the water pollution disrupt food chain in aquatic ecosystem?

Water pollution disrupts the food chain in aquatic ecosystems by harming organisms at different trophic levels and altering their interactions.

1. Death of Primary Producers

2. Bioaccumulation and Biomagnification

3. Decline of Primary Consumers

4. Collapse of Higher Trophic Levels

5. Habitat degradation

17.Write a short note on fixed cost in aquaculture

Fixed cost in aquaculture refers to the portion of production cost that does not change with the level of output. These expenses remain constant whether the farm produces a small amount of fish or operates at full capacity.

Incurred regardless of production volume

Repeated at regular intervals

Essential for maintaining farm infrastructure

Examples: Land lease or pond rent, Depreciation of ponds, tanks, cages, and equipment, Salaries of permanent staff, Interest on capital investment, administrative expenses

Importance

Fixed costs help determine the break-even point, guide financial planning, and assess the overall profitability of an aquaculture enterprise.

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